Christopher Spencer

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Senior Data Science and Analytics Engineering Leader with 15+ years of experience delivering production-grade Python, Spark, and cloud-native machine learning platforms across financial services, healthcare, and enterprise analytics environments. Proven track record managing and mentoring Data Scientist and Data Engineering teams, driving advanced analytics initiatives from ideation through deployment. Deep hands-on expertise in Databricks, Apache Spark, SQL-based data platforms, and modern ML techniques, with strong foundations in software architecture, API-driven systems, and Linux-based operations. Experienced in building scalable, containerized analytics solutions using Docker, Airflow, and cloud services across AWS, Azure, and GCP. Adept at translating complex analytical insights into clear, actionable outcomes for executive and non-technical stakeholders. Demonstrated success in consulting and client-facing roles, supporting proposal development, solution design, and cross-functional delivery. Entrepreneurial mindset with a passion for applying advanced analytics, machine learning, and data platforms to solve high-impact business and social problems at scale.

Work History

Senior AI/ML Engineer

Rho | *New York, NY* — *July 2021 - Current*

- Led and managed multiple teams of **Data Scientists and Analytics Engineers**, overseeing delivery of enterprise-scale **machine learning** and **advanced analytics** solutions supporting core business decisioning platforms.
- Architected and implemented large-scale **Databricks** and **Apache Spark** environments to process billions of records, enabling faster model development cycles and significantly improving analytical throughput.
- Designed end-to-end **Python-based ML pipelines** incorporating feature engineering, model training, validation, and deployment using **Spark ML**, improving prediction accuracy and operational reliability.
- Championed **software engineering best practices** within data science teams, enforcing **version control**, modular design, testing standards, and reusable **software architecture patterns**.
- Built and deployed **RESTful APIs** using **FastAPI** and **Flask** to operationalized machine learning models and analytical services for downstream applications and business consumers.
- Implemented **Docker-based containerization** and **Airflow orchestration** to standardize deployment and scheduling of analytics workflows across cloud environments.
- Partnered closely with business leaders to translate complex analytical outputs into clear insights, enabling data-driven decisions across risk, fraud, and customer intelligence initiatives.
- Collaborated with cloud engineering teams to deploy analytics workloads across **AWS** and **Azure**, optimizing performance, cost efficiency, and scalability.
- Established **Linux-based development and operational standards**, supporting secure, reliable execution of production analytics systems.
- Introduced **advanced machine learning techniques** including ensemble models, anomaly detection, and time-series forecasting to solve high-impact business problems.
- Supported consulting-style engagements with internal stakeholders, shaping problem statements, defining analytical approaches, and presenting results to executive audiences.
- Contributed to proposal development and solution design for new analytics initiatives, aligning technical capabilities with business objectives and delivery timelines.

- Mentored senior and junior team members, fostering a collaborative culture focused on continuous learning, innovation, and outcomes-driven delivery.
- Drove cross-functional collaboration across engineering, product, and compliance teams to ensure analytics solutions met regulatory, security, and operational requirements.

AI/ML Engineer

P3 Health Partners | *Henderson, NV* *March 2018 - July 2021*

- Led a team of **Data Scientists** delivering advanced analytics and **machine learning** solutions supporting healthcare operations, population health, and cost optimization initiatives.
- Designed and implemented scalable **Python** and **Apache Spark** data pipelines to process clinical and claims data across relational and **NoSQL databases**.
- Leveraged **Databricks** to accelerate model development, experimentation, and collaborative analytics across distributed teams.
- Developed predictive models using **modern ML techniques**, including risk stratification and utilization forecasting, improving operational planning and patient outcomes.
- Built and maintained **SQL-based analytics layers** using **PostgreSQL** and **SQL Server**, enabling reliable reporting and downstream analytical consumption.
- Created **API-driven services** using **Flask** and **Django** to expose analytics outputs to internal applications and care management platforms.
- Orchestrated analytics workflows with **Airflow**, ensuring reliable scheduling, monitoring, and recovery of critical data processing jobs.
- Deployed containerized analytics solutions using **Docker**, improving portability and consistency across development and production environments.
- Collaborated directly with healthcare leaders and non-technical stakeholders to translate analytical findings into actionable strategies and measurable outcomes.
- Supported client-facing and consulting-style engagements, presenting analytical insights and technical approaches to executive audiences.
- Contributed to solution design discussions, aligning analytical capabilities with evolving business and regulatory requirements.
- Mentored junior team members, reinforcing strong engineering discipline, documentation standards, and collaborative delivery practices.

Senior Software Engineer (Python/Full Stack)

JPMorgan Chase & Co | *Manhattan, NY* *February 2011 - March 2018*

- Built enterprise-grade **Python** and **Java** services for data-driven applications, focusing on reliability, security, and scalable integration patterns.
- Designed and implemented data integration pipelines across internal systems using **SQL** and messaging patterns, improving timeliness and consistency of critical datasets.
- Developed full-stack web applications with modern UI patterns, improving usability for operations and reducing time-to-complete for high-volume workflows.
- Implemented robust API layers with strict validation, **authN/authZ**, and audit logging aligned with financial regulatory requirements and internal risk controls.
- Optimized database performance through indexing, query refactoring, and schema improvements, reducing latency for high-traffic transactional workflows.
- Built reusable libraries for service bootstrapping, logging, and error handling, increasing developer velocity and standardizing quality.
- Implemented CI/CD pipelines and automated quality gates (tests, static analysis), reducing release risk and improving deployment frequency.

- Collaborated on event-driven designs leveraging messaging and asynchronous processing to decouple systems and improve throughput under peak load.
- Developed monitoring dashboards and alerting that improved anomaly detection and reduced mean time to detect production issues.
- Implemented secure secret handling and key management integrations, strengthening posture against credential leakage and misconfiguration.
- Supported data reconciliation and reporting workflows, improving traceability and accuracy for downstream consumers and auditors.
- Led design reviews, aligning stakeholders on architecture, non-functional requirements, and delivery milestones.
- Mentored junior engineers in clean architecture, testing, and debugging practices, improving maintainability.
- Reduced operational toil by automating recurring support tasks and improving runbooks, decreasing escalations and on-call interruptions.
- Delivered cross-team integrations through clear contracts, versioning, and backward compatibility to minimize breaking changes.

Skills

- **Languages:** Python, TypeScript, Java, C#, SQL, Bash, JavaScript, HTML/CSS
- **AI/ML:** LLMs, NLP, classification, recommendation, feature engineering, model evaluation, prompt engineering, RAG, vector search
- **Data Engineering:** Airflow, Spark, dbt, Kafka, Snowflake, Databricks, Delta Lake, Redshift, BigQuery
- **Backend & APIs:** FastAPI, Flask, Django, Node.js, Spring Boot, REST, GraphQL, gRPC, microservices
- **Frontend:** React, Next.js, Redux, TypeScript, Material UI, Tailwind, Jest, Cypress
- **Cloud & DevOps:** AWS (S3, EC2, Lambda, EKS, SageMaker), Azure (Functions, App Service, Key Vault), Docker, Kubernetes, Terraform, CI/CD
- **Observability & Security:** Prometheus, Grafana, CloudWatch, ELK, OpenTelemetry, OAuth2, OIDC, IAM, secrets management

Key Achivements

- **AI & Machine Learning Platforms:** Designed and deployed production-scale machine learning systems using Python, Spark, and modern ML frameworks to support fraud detection, anomaly detection, forecasting, and decision intelligence. Implemented feature engineering, model training, and validation pipelines that improved prediction accuracy and operational decision making across financial and healthcare domains.
- **Data Engineering & Distributed Processing:** Architected large-scale data processing platforms using Apache Spark, Databricks, and modern data warehouse technologies to process billions of records efficiently. Built robust data pipelines and orchestration workflows using Airflow and SQL-based platforms, significantly improving analytical throughput and reducing processing time for enterprise analytics workloads.
- **LLMs & Advanced Analytics:** Implemented advanced analytics solutions leveraging NLP techniques, vector search, RAG architectures, and prompt engineering to enable knowledge extraction, recommendation systems, and conversational analytics capabilities for internal decision support platforms.
- **API Platforms & ML Operationalization:** Built scalable API-driven architectures using Python-based services to operationalize machine learning models and analytics outputs. Developed secure and reliable service layers enabling real-time access to predictive insights for downstream business applications and internal data products.
- **Cloud & Scalable Infrastructure:** Designed and deployed containerized analytics platforms across AWS and Azure environments using Docker, Kubernetes, and infrastructure automation. Optimized distributed workloads for performance, cost efficiency, and reliability while supporting large-scale enterprise data environments.

- **MLOps & Workflow Automation:** Implemented automated machine learning lifecycle workflows including model training, validation, deployment, and monitoring. Established CI/CD practices and containerized environments to standardize deployment pipelines and accelerate analytics delivery across teams.
- **Enterprise Data Architecture:** Built modern data platforms integrating relational and distributed data stores including Snowflake, Redshift, BigQuery, and Delta Lake. Designed scalable data models and analytics layers supporting high-volume enterprise reporting, predictive modeling, and real-time insights.
- **Observability & Reliability:** Implemented monitoring and observability frameworks using Prometheus, Grafana, and distributed logging systems to track analytics pipeline health, detect anomalies, and reduce system downtime across production environments.
- **Leadership & Cross-Functional Delivery:** Led and mentored cross-functional teams of data scientists and engineers delivering high-impact analytics initiatives. Collaborated closely with business leaders, product teams, and stakeholders to translate complex analytical insights into actionable strategies and measurable outcomes.

Education

Master's Degree in Computer Science November 2007

Everest College at *Gahanna, OH*